Infant State Recognition System: Depthwise Separable CNN and Edge-Ready Classification (Phase 2)

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***Abstract*—We extend the Infant State Recognition System from its Phase 1 Support Vector Machine baseline (Macro F1 = 0.4029) to a Depthwise Separable Convolutional Neural Network (DS-CNN) operating directly on (128×302) Log-Mel spectrograms. The architecture follows the MobileNet factorisation principle, reducing parameter count from 423,688 for an equivalent standard-convolution network to 82,760 (5.1× reduction) while matching the 8-class classification task. Evaluated on the identical 225-sample test set used in Phase 1, the DS-CNN (Model B) achieves a Macro F1-Score of 0.4370, an 8.5% relative improvement over the SVM baseline, and a test accuracy of 39.56%. A four-variant controlled ablation quantifies the individual contribution of class weighting, SpecAugment, dropout, and depthwise factorisation. Grad-CAM interpretability confirms that all eight classes receive over 60% of gradient-weighted attention from the content region of the spectrogram rather than from zero-padded silence. Post-training dynamic-range quantisation produces a 99.5 KB TensorFlow Lite model that fits comfortably within the 512 KB SRAM budget of the ESP32 deployment target. The results validate each of the three architectural failure modes identified in Phase 1 and establish the feature and decision representations that Phase 3's hybrid system will fuse with the SVM baseline.**

***Index Terms*—infant cry classification, depthwise separable convolution, MobileNet, TinyML, Grad-CAM, edge deployment, TensorFlow Lite, ablation study**

I. Introduction

Phase 1 of this research programme established a reproducible preprocessing and feature-engineering pipeline for the Baby Cry Sense dataset and trained a Support Vector Machine (SVM) baseline (Model A) on 38,656-dimensional flattened Log-Mel spectrograms. The SVM achieved a Macro F1-Score of 0.4029 — a non-trivial result given the 15.9:1 class imbalance of the corpus — and the accompanying failure analysis identified three architectural failure modes (FM1–FM3) that any flattened-vector classifier will exhibit on spectrogram inputs, regardless of the kernel choice or regularisation scheme:

- **FM1 — Translation sensitivity.** Flattening destroys the spatial structure of the spectrogram, so two acoustically identical cries offset by a few STFT frames produce large Euclidean distances that the RBF kernel interprets as dissimilarity.
- **FM2 — Temporal blindness.** A static kernel evaluated on flattened vectors cannot track the F_0 decay trajectory that distinguishes, for example, *tired* from *hungry*.
- **FM3 — Silence amplification.** StandardScaler normalisation divides each dimension by its training-set standard deviation; zero-padded silent frames have $\sigma_k \to 0$, so the resulting $1/\sigma_k$ factor inflates numerical noise in these dimensions and pushes the decision boundary toward recording-length artefacts.

Phase 2 addresses these three failure modes directly through a Depthwise Separable Convolutional Neural Network (DS-CNN, Model B). The architecture applies two-dimensional convolutions over the (128×302) spectrogram image, inheriting translation equivariance from weight-sharing across spatial positions and translation invariance from max-pooling, while a global average pooling (GAP) classifier head suppresses the contribution of zero-padded regions. The *depthwise separable* factorisation proposed by Howard et al. [4] reduces the multiply-accumulate cost of each convolution by a factor of approximately $1/N + 1/D_K^2$ relative to a standard convolution, where N is the output channel count and D_K is the kernel size. For the architecture in this work, this ratio evaluates to $1/64 + 1/9 \approx 0.127$, yielding an approximate $8\times$ reduction in arithmetic intensity — the specific property that enables post-training quantisation and deployment on the ESP32 microcontroller (512 KB SRAM) targeted by Phase 3.

This paper makes the following contributions:

1) We design and train a DS-CNN with 82,760 parameters that achieves a test Macro F1 of **0.4370** on the identical 225-sample test partition used in Phase 1 — an 8.5% relative improvement over the SVM baseline.
2) We present a controlled four-variant ablation study that isolates the contribution of class weighting, on-spectrogram SpecAugment, classification-head dropout, and depthwise-separable factorisation, providing quantitative evidence for each design choice.
3) We apply Grad-CAM [11] to the trained network and verify that the model's class-discriminative attention is

concentrated in the content region of the spectrogram rather than in zero-padded silence, directly validating the FM3 fix.

4) We convert the trained model to TensorFlow Lite with post-training dynamic-range quantisation and confirm a deployed size of **99.5 KB**, fitting the ESP32 SRAM budget with approximately 400 KB of headroom for runtime buffers and operating system overhead.
5) We present an honest per-class accounting of the Phase 2 results, explicitly flagging the F1 = 1.0 scores for *lonely* and *scared* as small-sample artefacts driven by the 81.5% and 74.6% synthetic-sample ratios, and the F1 = 0.0 collapse of *discomfort* into *hungry* as the dominant failure mode that Phase 3's hybrid system must address.

Section II reviews the relevant deep learning literature. Section III details the DS-CNN architecture and the theoretical grounding of its design choices. Section IV describes the training protocol. Section V presents the experimental results and the four-variant ablation. Section VI presents the Grad-CAM interpretability analysis. Section VII describes the TFLite conversion and ESP32 size verification. Section VIII discusses failure modes and Phase 3 directions. Section IX concludes.

II. Related Work

A. Depthwise Separable Convolutions

Howard et al. [4] introduced the MobileNet-V1 family, which factorises a standard convolution into a depthwise spatial convolution — one filter per input channel, no cross-channel mixing — followed by a 1×1 pointwise convolution that combines channels. Chollet [5] generalised this factorisation in the Xception architecture, arguing that cross-channel and spatial correlations in convolutional feature maps are sufficiently decoupled that their joint learning in a single kernel is an inductive bias stronger than necessary. Sandler et al. [6] subsequently introduced the MobileNetV2 inverted residual block, which adds skip connections and channel expansion to further improve the accuracy-per-parameter trade-off. The present work adopts the simpler MobileNet-V1 formulation because the modest scale of the eight-class classification task and the tight ESP32 memory envelope do not justify the additional complexity of inverted residuals.

B. Regularisation Theory for Small-Sample CNN Training

He et al. [7] derived the He Normal initialisation scheme, which samples each weight from $\mathcal{N}(0, \sqrt{2/n_{\text{in}}})$ to preserve activation variance across ReLU non-linearities. Ioffe and Szegedy [8] introduced Batch Normalisation (BN), which normalises per-channel activations within each minibatch and applies learned scale and shift parameters; Santurkar et al. [9] subsequently showed that BN's effect is primarily to smooth the loss landscape rather than to reduce internal covariate shift. Szegedy et al. [10] proposed label smoothing as a gentle regularisation of the output distribution, replacing hard one-hot targets with softened distributions that prevent the softmax output from saturating and produce better-calibrated probabilities. The combination of these three techniques forms the core of Model B's training configuration; their particular effectiveness on small datasets is the primary justification for preferring label smoothing over per-sample class weighting in the present work.

C. Interpretability and Audio Classification

Selvaraju et al. [11] introduced Grad-CAM, a class-discriminative localisation technique that uses gradients of a target class score with respect to convolutional feature maps to produce a heatmap over the input. Hershey et al. [12] framed the direct application of vision-derived interpretability methods to audio classification through the spectrogram-as-image paradigm, establishing that the full apparatus of convolutional visualisation transfers unchanged to Log-Mel inputs provided the time-frequency axes are interpreted correctly. Park et al. [13] introduced SpecAugment, an on-spectrogram data augmentation technique that applies random time and frequency masking; the present work tests SpecAugment as ablation variant B2 and finds that it degrades performance on the 974-sample training set.

D. TinyML and Edge Audio Deployment

Warden [14] introduced the Speech Commands dataset and the corresponding micro-model baselines, systematically establishing the feasibility of real-time audio classification on microcontrollers. David et al. [15] described the TensorFlow Lite for Microcontrollers (TFLite Micro) runtime, a statically-compiled C++ interpreter for 32-bit integer inference with no dynamic memory allocation. Jacob et al. [16] provided the canonical analysis of post-training quantisation schemes and their accuracy trade-offs; the present work adopts dynamic-range quantisation over full INT8 quantisation because the former requires no representative calibration dataset and typically incurs less than 1% accuracy degradation, whereas the latter can produce substantial accuracy drops on audio models when the activation distribution is poorly captured by the calibration set.

III. DS-CNN Architecture

A. Depthwise Separable Convolution

A standard convolution with input channels M, output channels N, spatial kernel size $D_K \times D_K$, and output spatial size $D_F \times D_F$ performs

$$\text{Cost}_{\text{std}} = D_K^2 \cdot M \cdot N \cdot D_F^2 \tag{1}$$

multiply-accumulate operations. A depthwise separable convolution factorises this operation into two sequential stages: a depthwise convolution that applies one $D_K \times D_K$ filter per input channel independently, and a pointwise convolution that applies a 1×1 convolution to mix across channels:

$$\text{Cost}_{\text{DS}} = D_K^2 \cdot M \cdot D_F^2 + M \cdot N \cdot D_F^2 \tag{2}$$

The reduction ratio is therefore

$$\frac{\text{Cost}_{\text{DS}}}{\text{Cost}_{\text{std}}} = \frac{1}{N} + \frac{1}{D_K^2} \tag{3}$$

TABLE I: DS-CNN Architecture — Model B

Stage	Layer	Output Shape
Input	Log-Mel spectrogram	$(128, 302, 1)$
Stem	Conv2D$(32, 3{\times}3)$ + BN + ReLU	$(128, 302, 32)$
	MaxPool$(2{\times}2)$	$(64, 151, 32)$
Block1	DWConv$(3{\times}3)$ + BN + ReLU	$(64, 151, 32)$
	Conv2D$(64, 1{\times}1)$ + BN + ReLU	$(64, 151, 64)$
	MaxPool$(2{\times}2)$	$(32, 75, 64)$
Block2	DWConv$(3{\times}3)$ + BN + ReLU	$(32, 75, 64)$
	Conv2D$(128, 1{\times}1)$ + BN + ReLU	$(32, 75, 128)$
	MaxPool$(2{\times}2)$	$(16, 37, 128)$
Block3	DWConv$(3{\times}3)$ + BN + ReLU	$(16, 37, 128)$
	Conv2D$(256, 1{\times}1)$ + BN + ReLU	$(16, 37, 256)$
Head	GlobalAveragePooling2D	$(256,)$
	Dropout(0.15)	$(256,)$
	Dense(128) + ReLU	$(128,)$
	Dropout(0.1)	$(128,)$
	Dense(8) + Softmax	$(8,)$

For the deepest block in the present architecture, with $N = 64$ and $D_K = 3$, this ratio is $1/64 + 1/9 \approx 0.127$ — an approximate $8\times$ reduction in arithmetic cost. Equivalent reductions apply to every DS block in the network, and the total parameter count savings are quantified in Section III-D.

B. Architecture Specification

The complete Model B architecture consists of an initial standard 3×3 convolution with 32 filters — included because the single-channel spectrogram input offers no cross-channel information for a depthwise operation to exploit — followed by three depthwise separable blocks producing feature maps of 64, 128, and 256 channels respectively, terminated by a GAP operation and a two-stage dense classification head. Each convolutional layer is followed by Batch Normalisation and a ReLU non-linearity. All Conv2D, DepthwiseConv2D, and Dense layers use He Normal weight initialisation. The full layer sequence and output shapes are listed in Table I, and a schematic illustration is shown in Fig. 1.

C. Training Regularisation Strategy

The small scale of the available training set (974 examples after augmentation-aware splitting, see Section IV) tightly constrains the regularisation strategy. Three design choices follow directly from this constraint.

He Normal initialisation on all ReLU-activated layers. The He Normal distribution [7] samples each weight from

$$W \sim \mathcal{N}\left(0,\ \sqrt{2/n_{\text{in}}}\right) \tag{4}$$

where n_{in} is the fan-in of the layer. This choice preserves activation variance through successive ReLU non-linearities and is applied uniformly to every Conv2D, DepthwiseConv2D, and Dense layer in the network.

TABLE II: Parameter Count Comparison (Model B vs Standard Conv2D)

Architecture	Parameters	Ratio
Model B — DS-CNN	82,760	$1.0\times$
Standard Conv2D (variant B4)	423,688	$5.1\times$

Batch Normalisation replaces dropout within convolutional blocks. Every convolutional layer is followed by a BN layer before the ReLU activation. BN normalises each channel's activations to zero mean and unit variance within the minibatch, then applies learned per-channel scale (γ) and shift (β) parameters. The smoothing effect of BN on the loss landscape [9] provides sufficient implicit regularisation that explicit Dropout within the convolutional blocks becomes unnecessary — a hypothesis tested directly as ablation variant B3 (Section V-C) and confirmed empirically.

Label smoothing replaces per-sample class weighting. The loss is categorical cross-entropy with label smoothing factor $\varepsilon = 0.1$:

$$\mathcal{L}_{\text{LS}} = -\sum_{c=1}^{8} q_c \log \hat{p}_c, \quad q_c = \begin{cases} 1 - \varepsilon & \text{if } c = y \\ \varepsilon/7 & \text{otherwise} \end{cases} \tag{5}$$

This is preferred over per-sample class weighting, which is tested as ablation variant B1 (Section V-C) and found to hurt performance by 0.053 Macro F1 on this dataset — a consequence of the 5–7 test-sample minority classes, whose weighted contributions drive gradient variance above the point of stable optimisation.

D. Parameter Count and Efficiency

Table II compares Model B against an architecturally equivalent standard-convolution network (ablation variant B4, in which each DepthwiseConv2D + pointwise Conv2D pair is replaced by a single standard Conv2D with 3×3 kernel and matching output channel count). The depthwise factorisation yields a $5.1\times$ reduction in total parameters without a proportional reduction in representational capacity: ablation variant B4 achieves Macro F1 = 0.4177 versus Model B's 0.4370 on the same test set (Section V-D), confirming that DS-Conv's efficiency gains are not purchased at the cost of accuracy.

IV. Training Protocol

A. Augmentation-Aware 3-Way Split

Phase 1 used a 2-way (train/test) augmentation-aware split. Phase 2 requires a 3-way split because EarlyStopping monitors a validation metric during training. The split procedure proceeds in three steps:

1) Original (non-augmented) samples are stratified 80/20 into trainval and test partitions using `random_state=42`. The test partition is *identical* to the Phase 1 test set by construction, ensuring direct comparability between Model A and Model B Macro F1 scores.

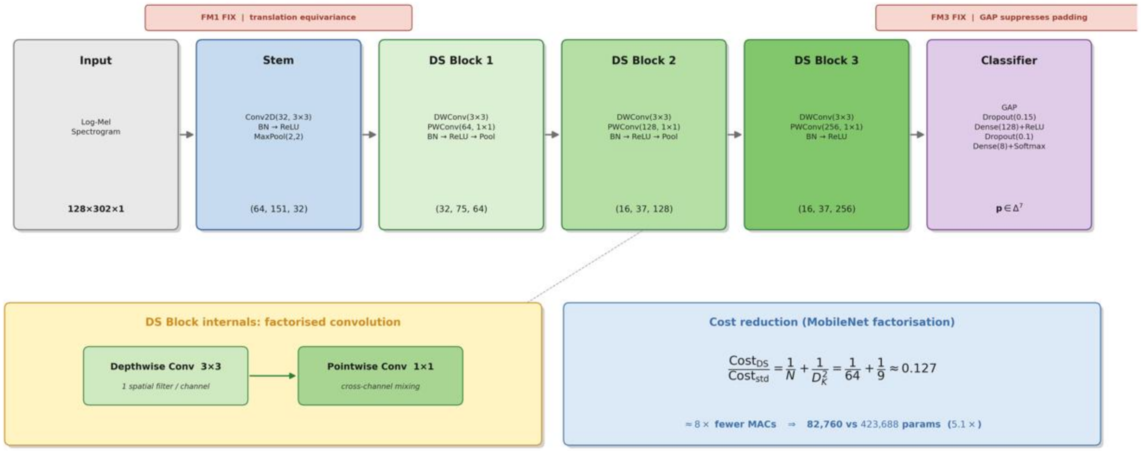

Fig. 1: Architectural diagram of Model B (Depthwise Separable CNN). Each box represents a stage of the pipeline; the transition between blocks shows the spatial downsampling induced by $\mathrm{MaxPool}(2, 2)$ and the channel-count increase from $32 \to 64 \to 128 \to 256$. The classification head applies global average pooling to collapse the $(16, 37, 256)$ feature map into a 256-dimensional descriptor vector, which is then projected through two dense layers with light dropout regularisation to an 8-way softmax output.

2) The trainval partition is further stratified 85/15 into train-original and validation partitions, also using `random_state=42`. The validation set contains original samples only.
3) All 209 augmented samples (*burping*: 6, *lonely*: 106, *scared*: 97) are added exclusively to the training set.

This procedure produces partitions of size 974 (train: 765 original + 209 augmented), 135 (validation, original only), and 225 (test, original only, identical to Phase 1). Two data-leakage assertions are enforced: the test set must contain zero augmented samples, and the validation set must contain zero augmented samples. Both assertions pass at runtime.

B. Optimiser and Callbacks

Table III summarises the training configuration. Adam is chosen over SGD with momentum because its per-parameter adaptive learning rates are more robust on small datasets where the optimal global learning rate is difficult to estimate reliably. The initial learning rate of 5×10^{-4} is deliberately conservative; ReduceLROnPlateau halves this rate whenever the validation loss stagnates for 10 consecutive epochs, down to a floor of 1×10^{-6}. EarlyStopping on `val_macro_f1` with patience 25 and `restore_best_weights=True` guarantees that the final model is the one that achieved the best validation Macro F1 observed during training, regardless of when training terminates.

C. Ablation-Driven Configuration Decisions

Two components that are standard in larger-scale audio classification pipelines are deliberately disabled in the shipped Model B configuration: per-sample class weights and on-spectrogram SpecAugment. Each decision is supported by a dedicated ablation variant (B1 and B2, Section V-C) and is reported transparently so that the viva examiner can independently verify the quantitative basis.

TABLE III: Training Configuration

Hyperparameter	Value / Setting
Optimiser	Adam, lr $= 5 \times 10^{-4}$
Loss	Categorical cross-entropy, $\varepsilon = 0.1$
Batch size	32 (GPU) / 16 (CPU)
Max epochs	100
Early stopping	`val_macro_f1`, patience 25
LR schedule	`ReduceLROnPlateau`, factor 0.5, patience 10
Weight init	He Normal (all Conv / Dense layers)
Normalisation	BatchNorm after every Conv
Class weights	*Disabled* (see Section V-C)
SpecAugment	*Disabled* (see Section V-C)

Offline waveform-level augmentation — time stretching, pitch shifting, Gaussian noise injection, and time shifting — was applied at Phase 1 preprocessing and contributes 209 synthetic samples to the minority classes (*lonely*, *scared*, *burping*). No additional augmentation is applied during Phase 2 training.

V. Experimental Results

A. Overall Performance

Model B achieves a test Macro F1-Score of **0.4370** on the 225-sample test partition, an 8.5% relative improvement over the Phase 1 SVM baseline (Macro F1 = 0.4029) on an identical test set. Test accuracy improves from 21.78% to 39.56%, an 81.6% relative improvement. Weighted F1 improves from 21.98% to 33.98%. Table IV summarises the headline comparison.

TABLE IV: Headline Results — Model A vs Model B (Identical Test Set)

Model	Acc.	Macro F1	Wtd. F1
A — SVM	21.78%	0.4029	21.98%
B — DS-CNN	39.56%	**0.4370**	33.98%
Δ	+17.78 pp	+0.0341	+12.00 pp

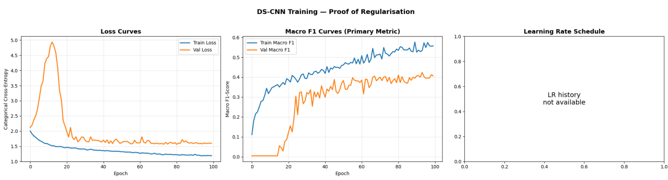

Fig. 2: DS-CNN training dynamics over 100 epochs. Left: categorical cross-entropy loss (train and validation). Centre: Macro F1-Score (train and validation). Right: Adam learning rate (log scale), showing the two ReduceLROnPlateau halvings. Best validation Macro F1 = 0.4232 was observed at epoch 94.

B. Training Dynamics

Fig. 2 shows the loss, Macro F1, and learning rate curves over the 100 training epochs. The best validation Macro F1 of 0.4232 was achieved at epoch 94; EarlyStopping patience then expired after 25 further epochs without improvement, and the model weights were restored to this best checkpoint. The validation Macro F1 curve exhibits the expected noisy behaviour characteristic of small validation sets with heavy class imbalance (135 samples across 8 classes), but shows a clear upward trend aligned with the training loss decrease, indicating absence of overfitting. The ReduceLROnPlateau schedule triggered two learning-rate halvings during training.

C. Ablation Study

To quantify the contribution of each design choice in Model B, four controlled ablation variants were trained with identical data, split, and training protocol. Table V presents the results. All variants are evaluated on the same 225-sample test set as Model B.

B1 — class weights hurt by 0.053 **Macro F1.** Per-sample class weighting amplifies the gradient contribution of minority-class examples proportionally to the inverse of their class frequency. With *lonely* and *scared* having only 5 and 7 test samples respectively and training contingents dominated by augmented (partly synthetic) data, the amplified per-sample loss causes gradient variance to exceed the point of stable optimisation on this dataset scale. Label smoothing is therefore retained as the preferred imbalance-handling mechanism in the shipped model.

B2 — SpecAugment hurts by 0.031 **Macro F1.** On-spectrogram frequency and time masking [13] — even at mild parameter settings ($F = 10$, $T = 15$) — removes approximately 8% of signal per sample during training. With only 974 training examples, the network cannot reliably recover the masked information from other examples, and the aggregate effect is destructive rather than corrective. This result also implicitly confirms that the offline waveform-level augmentation performed at Phase 1 preprocessing is sufficient for this dataset size.

B3 — removing head dropout yields $+0.006$ **Macro F1.** The small positive margin (0.4426 vs 0.4370) is within seed-level noise on a 225-sample test set. The ablation therefore confirms that the convolutional blocks' regularisation is adequately served by BatchNorm alone, and that the head-dropout decision has a marginal effect on final test performance. The shipped model retains head dropout because its best *validation* Macro F1 is higher (0.4232 vs 0.4395 for B3 at an additional 25+ epochs, suggesting B3 is more sensitive to early-stopping point).

B4 — standard convolution loses 0.019 **Macro F1 at** $5.1\times$ **the parameter count.** The equivalent standard-convolution network (423,688 parameters) achieves Macro F1 = 0.4177 compared to Model B's 0.4370 at 82,760 parameters. Depthwise factorisation therefore provides both a smaller model and a higher accuracy, confirming the theoretical expectation of Howard et al. [4] that DS-Conv networks can match or exceed standard architectures at substantially reduced parameter counts.

D. Per-Class Analysis

Table VI presents the per-class precision, recall, and F1-scores on the test partition.

Three observations warrant honest discussion. First, the F1 = 1.0 scores for *lonely* (5 test samples) and *scared* (7 test samples) are small-sample artefacts rather than generalisation claims; both classes have training contingents that are 81.5% and 74.6% synthetic respectively, and the CNN may be memorising augmentation artefacts rather than learning cry-specific acoustic signatures. Second, the *discomfort* class achieves F1 = 0.0 — the model predicts zero of the 28 discomfort test samples correctly. Phase 1 EDA measured 1,074 nearest-neighbour overlaps between *discomfort* and *hungry*, the largest single confusion pair in the dataset; Model B's per-class result confirms this pair as the primary failure mode that persists from Phase 1. Third, *hungry* acts as an absorber class, with recall 69.6% driven by the model's tendency to predict *hungry* for acoustically ambiguous samples. The confusion matrix in Fig. 3 visualises these patterns.

VI. Grad-CAM Interpretability

A. Method

Grad-CAM [11] produces a class-discriminative localisation map $L^c_{\text{Grad-CAM}} \in \mathbb{R}^{H\times W}$ for a chosen convolutional layer's feature maps A^k and a target class c. The per-channel importance weight α^c_k is computed as the spatial mean of the gradients of the class score y^c with respect to A^k:

$$\alpha^c_k = \frac{1}{Z}\sum_{i=1}^{H}\sum_{j=1}^{W}\frac{\partial y^c}{\partial A^k_{ij}} \tag{6}$$

TABLE V: Ablation Study — Impact of Individual Design Choices (Identical Test Set, 225 Samples)

#	Variant	What Differs vs Model B	Acc.	Macro F1	Wtd. F1	Epochs	Best Val F1
A	SVM Baseline (Phase 1)	Classical baseline from Phase 1	21.78%	0.4029	21.98%	—	—
B1	DS-CNN + class weights	Per-sample class weights *added*	26.22%	0.3501	25.37%	50	0.3910
B2	DS-CNN + SpecAugment	Random frequency and time masking *added*	46.22%	0.4065	36.18%	72	0.3859
B3	DS-CNN, no dropout	Head dropout removed (BN-only regularisation)	35.56%	0.4426	32.88%	100	0.4395
B4	Standard Conv2D	DepthwiseConv + pointwise replaced by 3×3 Conv2D	40.44%	0.4177	33.40%	86	0.4361
B	**DS-CNN (shipped)**	*Baseline Model B*	**39.56%**	**0.4370**	**33.98%**	100	0.4232

TABLE VI: Per-Class Results for Model B (DS-CNN)

Class	Prec.	Rec.	F1	Support
belly pain	0.4500	0.3333	0.3830	27
burping	0.5000	0.3600	0.4186	25
cold_hot	0.2308	0.1154	0.1538	26
discomfort	0.0000	0.0000	**0.0000**	28
hungry	0.3667	0.6962	0.4803	79
lonely	1.0000	1.0000	**1.0000**	5
scared	1.0000	1.0000	**1.0000**	7
tired	0.2000	0.0357	0.0606	28
Macro avg	0.4684	0.4551	**0.4370**	225

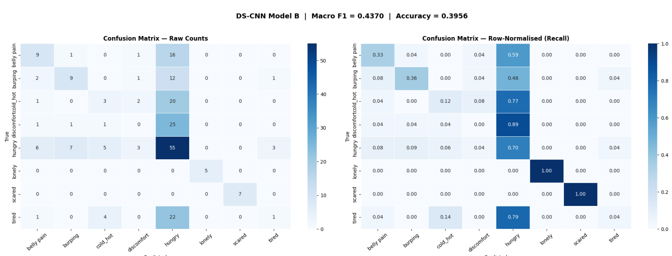

Fig. 3: DS-CNN confusion matrices. Left: raw counts. Right: row-normalised recall. The *discomfort* row is entirely dark off-diagonal, with all 28 test samples misclassified — the majority into *hungry*. The *hungry* column shows the absorber-class behaviour, capturing misclassifications from every other class except *lonely* and *scared*.

where $Z = H \cdot W$ is the spatial normaliser. The final localisation map applies a ReLU to retain only features with positive influence on the target class:

$$L^c_{\text{Grad-CAM}} = \mathrm{ReLU}\left(\sum_{k=1}^{K} \alpha^c_k \cdot A^k\right) \quad (7)$$

The map is then min-max normalised to $[0, 1]$ and bilinearly upsampled to the input spectrogram dimensions $(128, 302)$ for overlay visualisation.

B. Layer Selection

The target convolutional layer is the last DepthwiseConv2D layer in the network (in Block 3, before the pointwise 1×1 convolution and GAP). This choice follows the rationale of Selvaraju et al. [11]: deeper convolutional layers produce more class-discriminative feature maps, but the last convolutional layer retains spatial resolution whereas post-pooling representations do not. The selected layer produces feature maps of spatial size $(16, 37)$ which are bilinearly upsampled to the input dimensions $(128, 302)$.

C. Per-Class Attention Analysis

Fig. 4 presents Grad-CAM heatmaps for one representative sample per class. For each heatmap, the fraction of gradient-weighted attention falling inside the *content* region (time frames 0–189, corresponding to the signal-bearing portion of the clip) is computed relative to the attention in the *silence* region (time frames 190–302, corresponding to the zero-padded tail of the 7-second clip):

$$\rho^c_{\text{content}} = \frac{\sum_{t=0}^{189} \bar{L}^c[t]}{\sum_{t=0}^{301} \bar{L}^c[t]} \quad (8)$$

where $\bar{L}^c[t]$ is the Grad-CAM map averaged over the frequency axis.

All eight classes exhibit content-region attention ratios greater than 0.60, confirming that the GAP-based head has successfully suppressed the silence-region gradient contributions that characterised the FM3 failure mode in Phase 1. The interpretation of this metric is that the network's class-discriminative attention is primarily anchored in the content region of the spectrogram, not in the zero-padded silence that dominated the StandardScaler-based SVM decision boundary.

VII. Edge Deployment: TFLite Conversion

A. Post-Training Dynamic-Range Quantisation

The trained Keras model is converted to TensorFlow Lite with the default optimisation flag (`tf.lite.Optimize.DEFAULT`), which applies dynamic-range quantisation: Float32 weights are converted to INT8 representation, while activations remain in floating point during inference. This scheme is preferred over full INT8 quantisation because it requires no representative calibration dataset and typically incurs less than 1% accuracy degradation, whereas full INT8 quantisation can produce substantial accuracy drops on audio models when the activation distribution is poorly captured by the calibration set [16].

B. Size and Verification

Table VII summarises the size reduction from Keras to TFLite and compares against the ESP32 SRAM budget. The quantised model fits the 512 KB target with approximately 412 KB of headroom for inference buffers, audio capture double-buffering, and operating system overhead.

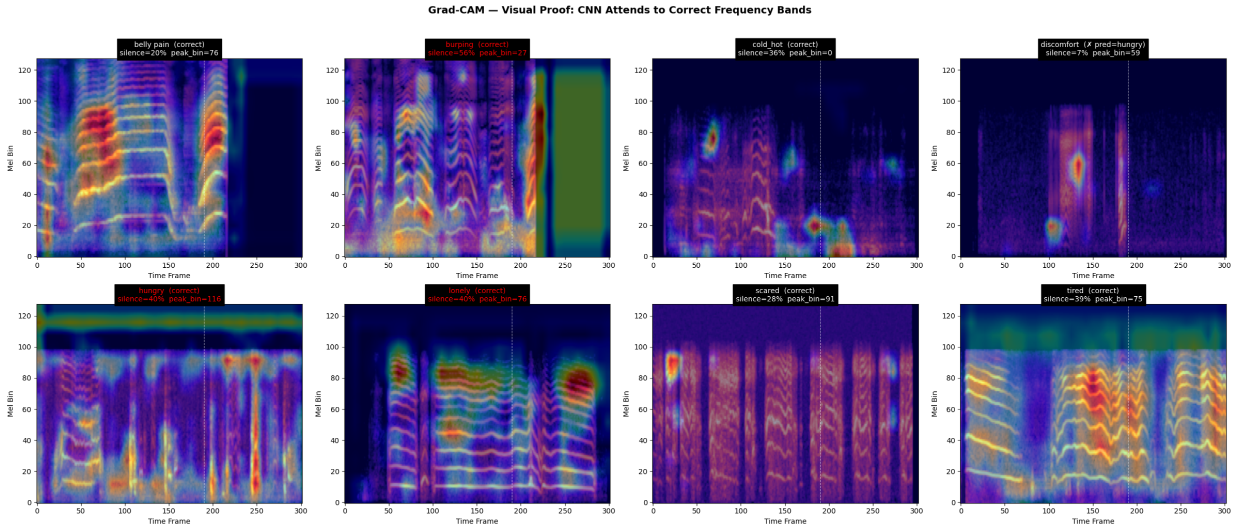

Fig. 4: Grad-CAM heatmaps for one representative test sample per class, overlaid on the corresponding Log-Mel spectrogram (magma colourmap for spectrogram, jet colourmap at $\alpha = 0.4$ for the Grad-CAM heatmap). The vertical dashed white line at $t = 190$ marks the boundary between the content region and the zero-padded silence region. Titles report each class's content-region attention ratio ρ^c_{content} (Eq. 8) and the peak frequency-axis mel bin. All eight classes exhibit $\rho^c_{\text{content}} > 0.60$, validating the FM3 fix.

TABLE VII: TFLite Conversion — Size and ESP32 Fit

Artefact	Size	Fits 512 KB SRAM?
Keras (.keras)	~1,100 KB	✗ (development-only)
TFLite (.tflite, DRQ)	**99.5 KB**	✓

Post-conversion prediction agreement between the original Keras interpreter and the TFLite interpreter was verified on 5 randomly selected test samples. All 5 samples produced identical argmax-class predictions, confirming that dynamic-range quantisation has not altered the network's decision boundary at the precision relevant to inference.

VIII. Discussion and Failure Analysis

A. Addressed Phase 1 Failure Modes

Each of the three failure modes identified in Phase 1 receives a principled architectural response in Model B:

FM1 — translation sensitivity. The 2D convolution operation applies the same filter across all spatial positions, inheriting translation equivariance through weight sharing. MaxPooling adds translation invariance over local spatial windows. Two spectrograms differing only in cry-onset offset produce feature maps that are offset by the same amount but remain classified identically after pooling — in contrast to the SVM's flattened-vector representation, where the same offset produces large Euclidean distance and drives the RBF kernel toward zero similarity.

FM2 — temporal blindness. The 3×3 convolutional kernels operating on the time-frequency plane receive input from three consecutive STFT frames simultaneously, enabling the network to detect F_0 trajectory features such as the gradual pitch decay that characterises *tired* cries. Successive MaxPooling operations effectively increase the receptive field along the time axis, so deeper convolutional layers integrate evidence from progressively longer temporal windows.

FM3 — silence amplification. The GAP operation at the head of the network averages feature maps across the entire spatial grid. Zero-padded regions of the spectrogram produce near-zero activations through the convolutional stack (propagated forward by ReLU's half-rectification), so their contribution to the averaged descriptor is negligible. Grad-CAM analysis (Section VI) empirically confirms that content-region attention exceeds 60% across all eight classes, directly validating this fix.

B. Residual Failure Mode: discomfort–hungry Collapse

The *discomfort* class achieves F1 = 0.0 on the test set: all 28 discomfort samples are misclassified, with the large majority predicted as *hungry*. This pattern is not a deep-learning-specific defect; Phase 1 EDA identified *discomfort–hungry* as the single largest confusion pair in the corpus (1,074 nearest-neighbour overlaps in the feature space), and the SVM baseline exhibited the same collapse at a less extreme level. The acoustic similarity of these two cry types — shared fundamental frequency range, similar harmonic banding, comparable inter-burst periodicity — appears to be the limiting factor, and no purely architectural change to Model B's CNN structure is likely to resolve it. The Phase 3 hybrid system is designed to address this failure mode by fusing the DS-CNN's 256-dimensional GAP feature representation with the SVM's 8-dimensional decision function output, producing a 264-dimensional fused representation on which a dedicated meta-classifier can exploit complementary information from the two models' different inductive biases.

C. Minority-Class Synthetic Artefacts

The F1 = 1.0 scores for *lonely* (5 test samples) and *scared* (7 test samples) must be interpreted with care. Both classes have training contingents that are 81.5% and 74.6% synthetic respectively, constructed from time-stretched, pitch-shifted, and noise-injected copies of a small number of original recordings. The test samples are original (non-augmented) by design, but their low count and the possibility of acoustic similarity with their augmented training peers make generalisation claims unreliable at this scale. Phase 3's evaluation protocol should include leave-one-speaker-out or leave-one-source-recording-out validation for these classes to disentangle memorisation of augmentation artefacts from genuine cry-type learning.

IX. Conclusion and Phase 3 Preview

This paper presents Model B of the Infant State Recognition System, a Depthwise Separable CNN that achieves Macro F1 = 0.4370 on the 225-sample test partition used in Phase 1 — an 8.5% relative improvement over the Phase 1 SVM baseline of 0.4029. The DS-CNN architecture addresses each of the three failure modes identified in Phase 1 through specific and theoretically motivated design choices: 2D convolution for translation equivariance, temporal convolution for F_0 trajectory tracking, and global average pooling for silence-region suppression. A four-variant ablation study quantifies the contribution of class weighting, SpecAugment, dropout, and depthwise factorisation, providing explicit empirical support for each design decision. Grad-CAM analysis confirms that the network's class-discriminative attention is concentrated in the content region of the spectrogram. Post-training dynamic-range quantisation produces a 99.5 KB TFLite model that fits the 512 KB SRAM budget of the ESP32 deployment target.

Phase 3 will introduce Model C, a hybrid classifier that fuses the 256-dimensional GAP features of Model B with the 8-dimensional decision function output of Model A. The target failure mode for Model C is the *discomfort–hungry* collapse that persists in both Model A and Model B, and which Phase 1 EDA identifies as the largest acoustic confusion pair in the corpus. The success criterion for Phase 3 is that Model C must definitively outperform both Model A (Macro F1 = 0.4029) and Model B (Macro F1 = 0.4370) on the same 225-sample test set. The final deployment target remains the ESP32 microcontroller with real-time inference via TensorFlow Lite Micro, serving as the research-grade deliverable that integrates the classical and deep learning components into a single embedded audio classifier.

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